

Kartik Rajendra Hirijaganer

Washington DC • (240) 854-7848 • kartikh@terpmail.umd.edu • [LinkedIn](#) • [GitHub](#) • [Portfolio](#)

WORK EXPERIENCE

Premier Health Group - Washington, DC | AI & Platform Engineer (Technical Lead)

January 2026 – Present

- Built a human-in-the-loop referral triage service that extracts coverage and clinical fields from payer portals, auto-routes low-risk cases, and escalates ambiguous ones with reason codes to care teams, cutting median decision time 15x, from 30 to 2 min.
- Shipped ambient documentation for 90-min behavioral health sessions using Whisper and GPT-4o structured extraction, with interruption filtering, PHI safeguards via LLM Guard, and clinician sign-off, cutting post-session charting by ~20 min.
- Architected a secure, tenant-aware AI execution layer using FastAPI, PostgreSQL, Redis, and Celery, with RBAC, audit trails, retry-safe asynchronous workflows, and human-approval gates for clinical use cases.
- Architected and coded Nightingale-OS, a multi-tenant healthcare platform built with FastAPI, React, Flutter, PostgreSQL, Redis, and Celery; led an 8-person engineering team through three releases in eight months using agile/scrum practices.

Premier Health Group - Washington, DC | AI Engineer Intern

June 2025 – December 2025

- Built a multimodal referral-ingestion service that extracts structured fields from scanned PDF referrals and prefills intake forms in under 90 seconds, attaching field-level confidence scores and human-review flags to prevent unsafe automation.
- Shipped a stateless, HIPAA-compliant X12 270/271 parser, encoder, and validation API for DC Medicaid eligibility that catches member, payer, and coverage mismatches before claim submission, contributing to a clean-claim rate increase from 74% to 94%.

Biz2Credit Info Services Private Limited - Mumbai | Data Engineer

January 2023 – July 2024

- Engineered an Airflow/dbt/Snowflake reporting pipeline for HSBC processing 10K+ daily loan decisions, modeling approval, risk, and turnaround metrics into BI dashboards and eliminating ~15 hours of manual reporting per week.
- Developed streaming and batch ingestion for bank statements, credit-bureau feeds, and financial APIs, processing \$20M+ in daily transaction value at under 5-minute data freshness for downstream ML models.

Capgemini Technology Services India Limited - Mumbai | Software Engineer

September 2020 – January 2023

- Owned end-to-end the serverless integration connecting Subway and ezCater across 10,000+ store locations, using Node.js/TypeScript and an Apollo GraphQL API layer on a distributed, event-driven architecture.
 - Analyzed 50M+ production log events to trace a 4% store-initiated cancellation rate to stale inventory, then implemented pre-order validation and event-driven inventory sync, reducing cancellations to under 1%, a 75%+ relative reduction.
-

PROJECT EXPERIENCE

FraudLens – AML Fraud Investigation Platform | [GitHub](#)

- Implemented an explainable AML investigation workflow on LangGraph chaining deterministic rules, XGBoost/SHAP risk scoring, and citation-grounded FinCEN/BSA retrieval (ChromaDB) to draft auditable SAR narratives for human review.
- Deployed FraudLens microservices to Amazon EKS, defining Kubernetes Deployments, Services, and ConfigMaps on Terraform-provisioned AWS infrastructure.

Briefed - LLM Powered Email Triage App | [Live Demo](#)

- Shipped a multi-account Gmail triage product with scheduled SQS fan-out, schema-constrained LLM classification and summarization, provider fallback (Gemini 2.0 Flash to Claude Haiku), user-configurable sender and topic rules.
 - Deployed FastAPI services on AWS Lambda, CloudFront/WAF, and Supabase/PostgreSQL with KMS envelope encryption.
-

TECHNICAL SKILLS

Languages & Backend: Python, SQL, Bash, FastAPI, REST APIs, Celery

Generative AI & Agents: LLMs, RAG, LangGraph, MCP, Prompt Engineering, Structured Outputs

Machine Learning & Evaluation: PyTorch, TensorFlow, scikit-learn, XGBoost, SHAP, LLM Evaluation

Retrieval, Data & Streaming: Embeddings, Vector Databases (ChromaDB), PostgreSQL, Redis, Apache Kafka, Snowflake, Airflow, dbt

Cloud & MLOps: AWS (Bedrock, SageMaker, Lambda, S3, SQS), Kubernetes (EKS), Docker, Terraform, GitHub Actions, CI/CD

Safety & Reliability: Human-in-the-Loop, Confidence Scoring, Guardrails, RBAC, Audit Logging

EDUCATION & CERTIFICATIONS

Master of Science in Information Systems | University of Maryland - College Park, USA | GPA: 3.92

December 2025

Bachelor of Engineering | University of Mumbai, India | GPA: 7.52

June 2020

AWS Certified Data Engineer – Associate

January 2025

Azure AI Apps and Agents Developer - Associate (AI-103)

In Progress